

Confidence interval for difference of two means, dependent samples

Magnesium example

Patient	Before	After	Difference
1	2.00	1.70	-0.30
2	1.40	1.70	0.30
3	1.30	1.80	0.50
4	1.10	1.30	0.20
5	1.80	1.70	-0.10
6	1.60	1.50	-0.10
7	1.50	1.60	0.10
8	0.70	1.70	1.00
9	0.90	1.70	0.80
10	1.50	2.40	0.90

Mean 0.33
St. deviation 0.45

95% CI, $t_{9,0.025}$ 2.26

Confidence intervals

T	CI low	CI high
95%	0.01	0.65

*CI low stands for lower bound of the confidence interval
CI high stands for higher bound of the confidence interval
The percentages indicate the confidence
T stands for the fact that we are using the t-statistic
Note that this is not a common way to summarize the data.
It is just a clear and useful one in Excel

2. The Math Formula (using the t-distribution)

$$CI = \bar{x} \pm \left(t^* \cdot \frac{s}{\sqrt{n}} \right)$$

- $\bar{x}$ (x-bar): The **mean** (average) of your sample.
- t^* : The **critical value**. This comes from a t-table based on how confident you want to be (like 95%) and your "degrees of freedom."
- s : The **standard deviation** of your sample (how spread out the numbers are).
- n : The **sample size** (how many items or people you measured).
- $\frac{s}{\sqrt{n}}$: This whole part is called the **Standard Error**.

$$(\bar{x}_1 - \bar{x}_2) \pm (\text{Critical Value}) \times (\text{Standard Error})$$

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$\text{Difference} = \bar{x}_1 - \bar{x}_2$$

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Add and subtract the Margin of Error from your point estimate to get your lower and upper bounds.